

INVESTIGATION REPORT

Issue 1023: Two-cycle APU write-back path in the EX stage

openhwgroup/cv32e40p

Document	RTL-TRIAGE-CV32E40P-1023
Disposition	Dead code, proven
Record date	2026-09-17
Classification	Internal working document. Not a verification sign-off.

1 Status

Issue	openhwgroup/cv32e40p#1023
Issue filed	2024-07-10
Issue type	Coverage hole referred for disposition
Affects	cv32e40p
Configuration	RTL 6033d2b, COREV_PULP=1 FPU=1 (the upstream PULP_F0 configuration), top cv32e40p_formal_top, yosys-slang frontend
Disposition	Dead code, proven
Severity	Not assessed
Agreement	Overlaps the human record but disagrees on part of it.
Evidence	4 cited, 4 present

Disposition meaning. The code cannot be reached in any state permitted by the stated model and assumptions, established by an unbounded proof.

Severity is not recorded for this run. The harness does not infer a severity from the disposition, so the field is reported as not assessed rather than filled in.

2 Description

The uncovered branches at cv32e40p_ex_stage.sv:237 and 241 are unreachable in this configuration. Both lines carry the identical guard `apu_valid & (!apu_singlecycle & !apu_multicycle)`, the path that writes a two-cycle APU result back on the load/store port. At FPU latency 0 every APU operation is classified single cycle, so a valid APU result with neither classification set cannot occur. An unbounded proof (`abc pdr, top cv32e40p_formal_top, COREV_PULP=1 FPU=1, the upstream PULP_F0 configuration`) passes. Two bounded cover runs show the APU is alive in the model, reaching a valid single-cycle result at step 12 and a valid multi-cycle result at step 14, so the assertion is not vacuous. One difference from the upstream record: the published `assert_unreachable_ex_237` reads `regfile_we_lsu |-> ! (~apu_valid & (!apu_singlecycle & !apu_multicycle))`, which negates `apu_valid` where the RTL guard asserts it. This investigation states the guard as it appears in the RTL. The latency 2 configuration named in the issue was not run.

3 Conditions

1. legal OBI handshakes on both the instruction and the data interface
2. `scan_cg_en_i=0`
3. no CSR write to the hardware-loop CSRs (0xCC0, 0xCC1, 0xCC2, 0xCC4, 0xCC5, 0xCC6)
4. clock gates modelled as transparent
5. every trace starts in reset
6. instruction memory contents otherwise unconstrained

Reproduced from the assumes that are enabled in the upstream formal environment at `cores/cv32e40p/scripts/formal/src`, plus the two modelling choices the open-source flow requires.

4 Implications

Not recorded as a separate field for this run. What the disposition means for a user of the core follows from the Disposition meaning in Status and the bounds in Limitations and scope.

5 Action

Waiver text, to be applied to the coverage exclusion

cv32e40p_ex_stage.sv:237 and 241 (two-cycle APU write-back on the load/store port): the guard `apu_valid & (!apu_singlecycle & !apu_multicycle)` is formally proven never true (abc pdr, unbounded, top cv32e40p_formal_top, COREV_PULP=1 FPU=1 at latency 0, RTL 6033d2b). At that latency every APU operation is classified single cycle. Assumptions: legal OBI handshakes on both interfaces, `scan_cg_en_i=0`, no CSR write to the hardware-loop CSRs, transparent clock gates. Evidence: runs/v1-cv32e40p-1023/formal/p_assert/sby.log (PASS) and formal/c_antecedent/sby.log (2/2 APU classifications reachable). This waiver covers latency 0 only; the latency 2 configuration named in the issue has not been checked.

Checker used

checker_1023.sv

6 Evidence

Ref	Type	Path, relative to the run directory	File	Result recorded
E1	proof	formal/p_assert/sby.log	present	abc pdr PASS (COREV_PULP=1 FPU=1 latency 0): <code>apu_valid</code> is never set with both <code>apu_singlecycle</code> and <code>apu_multicycle</code> low
E2	proof	formal/p_assert/result.json	present	machine-readable status record for the proof job
E3	proof	formal/c_antecedent/sby.log	present	smtbmc/yices cover PASS (depth 60): <code>covers_reached=2</code> - a valid single-cycle APU result at step 12 and a valid multi-cycle result at step 14, so the APU is exercised and <code>a_ex237_241</code> is not vacuous
E4	proof	formal/c_antecedent/result.json	present	machine-readable status record for the cover job

Paths are relative to runs/v1-cv32e40p-1023. The File column is a check made when this document was generated. It reports whether the cited file exists, not whether its contents were re-validated. 'not kept' means a repository rule deliberately excludes that artefact, usually a waveform dump; rerun the check to regenerate it. 'MISSING' means the file is absent and was expected to be there.

What each class of artefact establishes

- proof. Proof or engine run. Scope depends on the mode: prove is unbounded under the stated model, bmc is bounded to its depth, cover establishes reachability only.

7 Limitations and scope

- The proof establishes unreachability under the model and the environment assumptions named in Status and Conditions. It does not establish that the surrounding logic is correct.
- This is an internal investigation record. It requires engineering review before it is used as a disposition. No report generator can supply missing evidence.
- Results apply to the configuration and core revision named in Status. A different configuration needs a separate run.
- Public demonstration runs on open cores are not client engagements and not production verification results.

8 Provenance and measurement

Run directory runs/v1-cv32e40p-1023
Inputs shown title, body(links stripped), labels
Human record Dolphin Design 2024: waived in v2; reported proven unreachable with Siemens Questa Static formal, assertion `assert_unreachable_ex_237`

Assessed confidence	0.88. This is the investigator's own estimate. It is not a measurement and not a reliability figure.
Iterations	1

Generated by reporting/scripts/issue_document.py from runs/v1-cv32e40p-1023/verdict.json. The generator typesets recorded data and checks that cited files exist. It does not run, rerun or validate any check, and it cannot supply missing evidence.